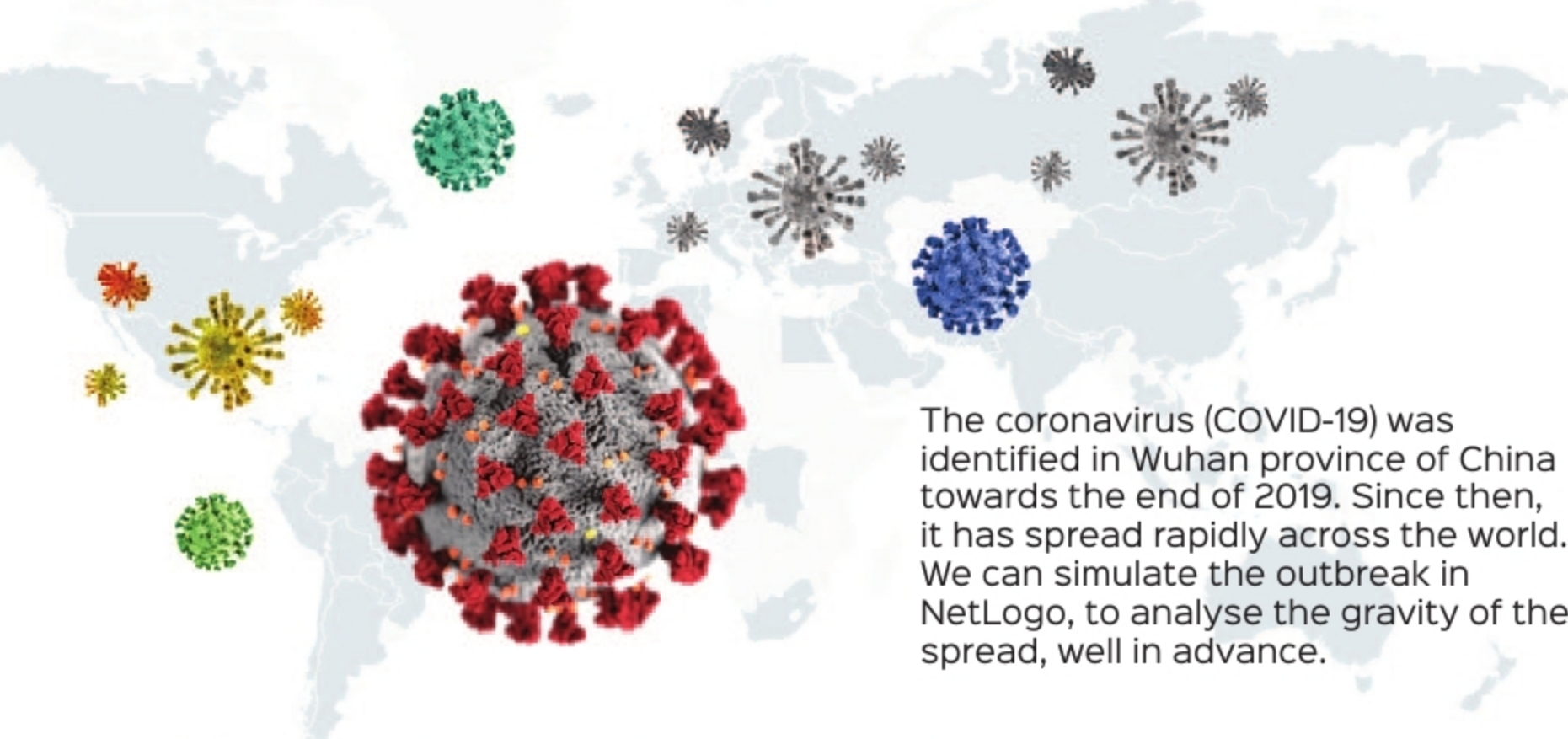


Simulating the Coronavirus Outbreak in Netlogo



The coronavirus (COVID-19) was identified in Wuhan province of China towards the end of 2019. Since then, it has spread rapidly across the world. We can simulate the outbreak in NetLogo, to analyse the gravity of the spread, well in advance.

According to different studies, the coronavirus can survive on a surface for up to 72 hours. This has quite literally locked down the world. The virus, which can spread through person to person contact, has infected over a million (as of the first week of April) and led to the deaths of thousands around the world. Even though modern medicine has advanced a lot, we have no antidote for such new viral diseases. We can't even predict how long developing a vaccine might take, and whether that may be too late for many.

To be well prepared for such viral outbreaks, it is important to know how the disease will spread, well in advance. Each country has the data on its population, density, demographics, etc. It is possible to simulate an outbreak for

different communities/countries if we quantify the parameters properly in a simulation. Netlogo is software that can simulate such scenarios on a large scale. It is GNU GPL licensed free software, designed by Uri Wilensky, who was the director of Northwestern University in 1999. It is a JVM based cross-platform multi-agent simulation platform.

NetLogo doesn't require coding knowledge or any other prerequisites. It is designed for a broad audience. So, anybody from secondary school students to graduates can make use of this software for their field of interest. An inbuilt model library has various models available on disease spread. Some are explained below.

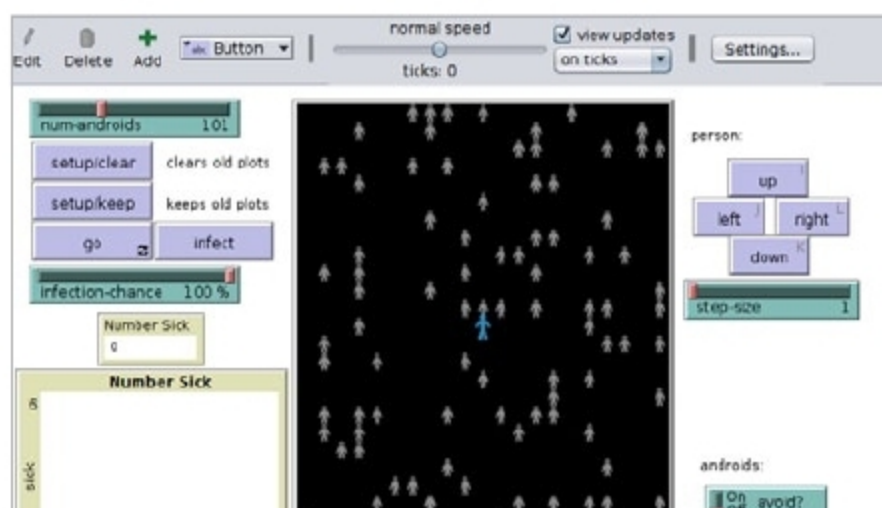


Figure 1: Disease Solo – initial situation

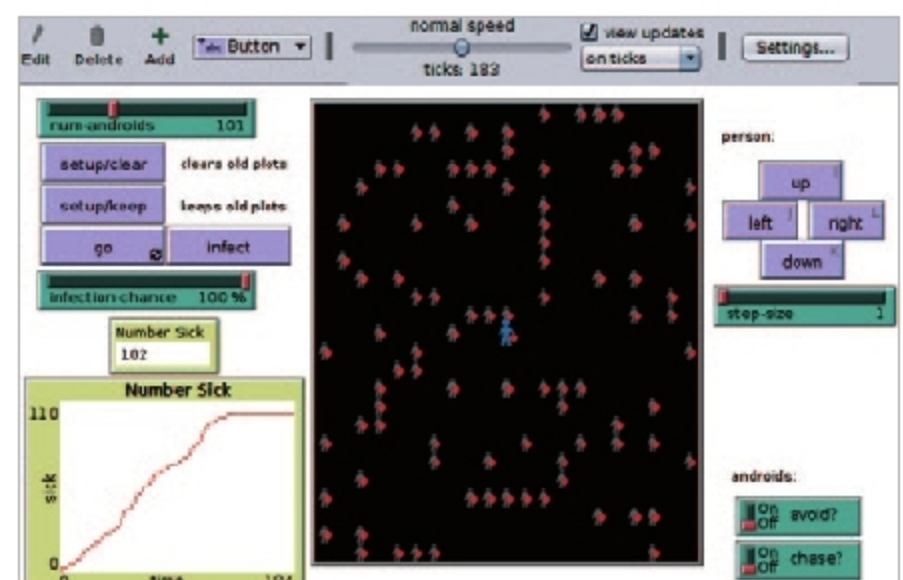


Figure 2: Disease Solo – final situation

